

Total No. of Questions : 8]

SEAT No. :

P484

[Total No. of Pages : 2

[6003]-704

T.E. (Information Technology)
HUMAN COMPUTER INTERACTION INFORMATION
TECHNOLOGY

(2019 Pattern) (Semester - I) (314444)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answers : Q.1. or Q.2, Q.3 or Q.4,, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume Suitable data if necessary.*

- Q1)** a) What are User Profiles explain with example? [8]
b) Hierarchical task analysis (HTA) is used to describe the interactions between a user & software system. Draw & explain HTA to online bus reservation system? [9]

OR

- Q2)** a) Explain Norman's 7 stage model with example. [8]
b) Differentiate between Goals Vs Tasks. How does GOMS help in improving usability? Draw GOMS for CLOSE-WINDOW. [9]

- Q3)** a) Explain the principle of learnability and principle of flexibility with example. [9]
b) i) Draw and explain Software design process. [9]
ii) Explain the importance of User Focus from HCI perspective.

OR

- Q4)** a) What is Prototyping? Explain the low-fidelity and High-fidelity designs with example. [9]
b) Consider any online food ordering system, draw model-view-controller (MVC) framework. Mention the necessary technology solutions available for each of MVC. [9]

- Q5)** a) Explain DECIDE framework with necessary diagram and an example of the same. [8]
b) What are the goals of evaluation? Explain Cognitive walkthrough and heuristics evaluation technique in detail. [9]

OR

P.T.O.

- Q6)** a) Explain user interface management system (UIMS) in detail along with its architecture? [8]
b) Write a short note on. [9]
i) Toolkits
ii) Usability testing

- Q7)** a) Explain [9]
i) Augmented Reality
ii) Virtual Reality along with real life examples of both.
b) Discuss in the detail the Challenges faced by designer while designing interfaces for [9]
i) smart homes
ii) smart devices

OR

- Q8)** a) Draw and explain Design thinking in detail for any suitable application.[9]
b) In today's world finding things on the web has become easy. Discuss how multimodal interaction has enriched the experience. [9]

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Total No. of Questions : 8]

SEAT No. :

PA-1502

[Total No. of Pages : 2

[5926]-122

T.E. (E&TC)

INFORMATION TECHNOLOGY

Human Computer Interaction

(2019 Pattern) (Semester - I) (314444)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answers : Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Explain Goal and task hierarchy model with example. [8]

- b) Hierarchical task analysis (HTA) is used to describe the interactions between a user & software system. Draw & explain HTA to online Movie booking system? [9]

OR

Q2) a) Differentiate User Profiles with respect to Interface design with example. [8]

- b) How does Diagrammatic dialog design notations help designers to design better interfaces. Justify your answer with an example. [9]

Q3) a) Explain the following golden rules with example. [9]

- i) Strive for consistency
 - ii) Enable frequent users to use shortcuts
 - iii) Offer informative feedback
- b) Explain the following with reference to interface design with example [9]
- i) Scenarios
 - ii) Navigation Design
 - iii) Screen Design

OR

P.T.O.

- Q4) a)** What is Prototyping? Explain the low-fidelity and High-fidelity designs with example. [9]
- b)** Consider any online digital library, draw Model -View- Controller (MVC) framework. Mention the necessary technology solutions available for each of MVC. [9]

- Q5) a)** What are the goals of evaluation? Explain Cognitive walkthrough and heuristics evaluation technique in detail. [8]
- b)** What is Usability testing? How will you perform Usability testing on an interactive interface? [9]

OR

- Q6) a)** Explain user interface management system (UIMS) in detail along with its architecture? [8]
- b)** Explain DECIDE framework with necessary diagram and an example of the same. [9]

- Q7) a)** Explain: i) Augmented Reality ii) Virtual Reality along with real life examples of both. [9]
- b)** Discuss in the detail the Challenges faced by designer while designing interfaces for [9]
- i) Smart homes ii) Smart devices

OR

- Q8) a)** Draw and explain Design thinking in detail for any suitable application. [9]
- b)** In today's world finding things on web has become very easy. Discuss how the multimodal input has enriched the experience. [9]



Total No. of Questions : 8]

SEAT No. :

P-7620

[Total No. of Pages : 2

[6180]-140

T.E. (Information Technology)
HUMAN COMPUTER INTERACTION
(2019 Pattern) (Semester - I) (314444)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) User Profiles are very important from design point of view. Interface with such profiles is helpful. Justify your answer? [8]
- b) Hierarchical task analysis (HTA) is used to describe the interactions between a user & software system. Draw & explain HTA to online bus reservation system? [9]

OR

- Q2)** a) Explain Norman's 7 stage model with example. [8]
- b) How does GOMS help in improving usability? Draw GOMS for CLOSE-WINDOW. Differentiate between Goals Vs Tasks. [9]

- Q3)** a) Explain the following principles with example. [9]
- i) Learnability
 - ii) Flexibility
- b) i) Draw and explain Software design process.
- ii) Explain the importance of User Focus from HCI perspective. [9]

OR

- Q4)** a) What is Prototyping? Explain the low-fidelity and High-fidelity designs with example. [9]
- b) Consider any online food ordering system, draw Model -View- Controller (MVC) framework. Mention the necessary technology solutions available for each of MVC. [9]

P.T.O.

Q5) a) Explain DECIDE framework with necessary diagram and an example of the same. [8]

b) What are the goals of evaluation? Explain Cognitive walkthrough and heuristics evaluation technique in detail. [9]

OR

Q6) a) Explain user interface management system (UIMS) in detail along with its architecture? [8]

b) Write a short note on-i) Toolkits ii) Usability testing [9]

Q7) a) Explain [9]

i) Augmented Reality

ii) Virtual Reality along with real life examples of both.

b) Discuss in the detail the Challenges faced by designer while designing interfaces for [9]

i) smart homes

ii) smart devices

OR

Q8) a) Draw and explain Design thinking in detail for any suitable application.[9]

b) In today's world finding things on web has become very easy. Discuss how the multimodal interaction has enriched the experience. [9]



Total No. of Questions : 8]

SEAT No. :

PB-3861

[Total No. of Pages : 2

[6262]-124

T.E. (Information Technology)
HUMAN COMPUTER INTERACTION
(2019 Pattern) (Semester - I) (314444)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) **Answers: Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.**
- 2) **Neat diagrams must be drawn wherever necessary.**
- 3) **Figures to the right indicate full marks.**
- 4) **Assume Suitable data if necessary.**

Q1) a) What are Norman's seven principles for reducing complex jobs to simple one? **[8]**

b) How does making a call differ when using: **[9]**

- i) Smart Cellphone
- ii) Land line phone

How have been these devices been designed to consider the following:

- i) The kind of users
- ii) Types of activity being supported
- iii) Contexts of use

OR

Q2) a) What exactly are mental models and why do they matter in interface design? **[5]**

b) Explain Low-fidelity and High-fidelity prototypes. **[6]**

c) Write a short note on the following: **[6]**

- i) Linguistic Model
- ii) Physical and device Model

P.T.O.

Q3) a) Consider two different ATM machines, one giving away the cash & then ejecting the bank card & the other one ejecting the bank card first & then dispensing the cash. Which is a better interface from interaction design point of view? justify. [9]

b) Explain the principle of Learnability and principle of Flexibility with relevant examples. [9]

OR

Q4) a) Write a Scenario for making an online transaction using any mobile phone-based application. [6]

b) What is design? What is the golden rule of design? Draw the process of interaction design. [6]

c) Draw and explain Model -View- Controller (MVC) framework for any suitable application. [6]

Q5) a) Cognitive Walk through is one of the important techniques with reference to HCI. Justify the statement with a suitable example of your own. [8]

b) Draw and explain User Interface Management System architecture. [9]

OR

Q6) a) How will you apply the DECIDE framework to any online booking system? Explain it with necessary diagram and suitable explanation. [10]

b) What do you mean by evaluation? What are the goals of evaluation? Explain any one category of evaluation in detail. [7]

Q7) a) What do you mean by design thinking? Explain its 5 stages in detail? [9]

b) Differentiate between smart devices and handheld devices. [5]

c) Mention your opinion regarding the future of HCI with an example. [4]

OR

Q8) a) Define the term Ubiquitous computing? Explain it in detail with reference to some suitable examples. [9]

b) Mention a real life example of Augmented Reality and Virtual Reality. [5]

c) What challenges will be faced by designer while designing interfaces for smart wrist bands. [4]



Total No. of Questions : 8]

SEAT No. :

PC-1804

[Total No. of Pages : 2

[6353] - 123

T.E. (I.T.)

HUMAN COMPUTER INTERACTION

(2019 Pattern) (Semester - I) (314444)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answers Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Categorize user profiles for any social media application based on usage. [8]

b) Hierarchical task analysis (HTA) is used to describe the interactions between a user & software system. Draw & explain HTA to online bus reservation system? [9]

OR

Q2) a) Consider designing a user interface for “on-line video store” or “student admission system”. Perform a detailed task analysis. Identify task domain object & actions. Relate task domain & actions with interface objects & actions. [9]

b) Create a GOMS description of the task of photocopying a paper from a journal. Discuss the issue of closure in terms of your GOMS description? [8]

Q3) a) What is a prototype? Explain low-fidelity, medium-fidelity and high-fidelity prototyping technique with relevant example. [9]

b) Explain the principle of Learnability and principle of Flexibility with relevant examples. [9]

OR

P.T.O.

Q4) a) Write short notes on: [9]

- i) Wire-Framing
- ii) Model-View-Controller (MVC) Framework

b) Consider two different ATM machines. One giving away the cash & then ejecting the bank card & the other one ejecting the bank card first & then dispensing the cash. Which is a better interface from interaction design point of view? justify. [9]

Q5) a) List & explain the steps of usability testing? What are some of the limitations of such testing? [8]

b) Explain user interface management system (UIMS) with its architecture? [9]

OR

Q6) a) What are the goals of evaluation? Explain Cognitive Walkthrough and heuristics evaluation technique in detail. [8]

b) Explain DECIDE framework with necessary diagram and an example of the same. [9]

Q7) a) Draw and explain the design thinking process. [10]

b) Discuss in the detail the Challenges faced by designer while designing interfaces for [8]

- i) Smart homes
- ii) Smart devices

OR

Q8) a) Differentiate Augmented and Virtual Reality. Also explain both with real life examples. [8]

b) Explain Ubiquitous Computing Definition, key features, Applications, Advantages & disadvantages, [10]



Total No. of Questions : 8]

SEAT No. :

PD4325

[Total No. of Pages : 2

[6403]-123

T.E. (Information Technology)
HUMAN COMPUTER INTERACTION
(2019 Pattern) (Semester - V) (314444)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat circuit diagrams should must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1) a)** Categorize user profiles for any mobile phone-based money transfer application. (Hint:- Google pay/ PhonePe/ BHIM, etc.) [7]
- b)** Goals are accomplished by methods consisting of operators which are identified by selection rules. illustrate this for following goals. [10]
- i) To delete a sentence in a graphical text editor
 - ii) To close window in a graphical text editor

OR

- Q2) a)** Create a GOMS description of the task of photocopying a paper from a journal. Discuss the issue of closure in terms of your GOMS description? [7]
- b)** Consider designing a user interface for “on-line movie ticket booking” or “hospital admission system”. Perform a detailed task analysis. Identify task domain object & actions. Relate task domain & actions with interface objects & actions. [10]

- Q3) a)** A scenario is not only an idealized but also a detailed description of a specific instance of HCI. Scenarios specify how users carry out their tasks in a specified context. Write scenarios for booking an airline ticket from any online application. [9]

Note: Generate scenarios to cover a wide range of situations. Include problem situations that will test the system concept.

- b)** Draw and explain Software design process from HCI perspective. [9]

OR

P.T.O.

- Q4) a)** Specify an example for low-fidelity, medium-fidelity, and High-fidelity prototype for any “Exam registration portal”. **[9]**
- b)** Consider any “Online grocery ordering system”, draw Model -View-Controller (MVC) framework. Mention the necessary technology solutions available for each of MVC. **[9]**

- Q5) a)** Explain DECIDE framework with necessary diagram and an example of the same. **[9]**
- b)** Explain user interface management system (UIMS) with its architecture? **[9]**

OR

- Q6) a)** What is Usability testing? How will you perform Usability testing on an interactive interface which deals into “Hotel booking”? **[9]**
- b)** Write a short note on- **[9]**
- i) Heuristic Evaluation
 - ii) Cognitive walk through

- Q7) a)** Explain the following along with real life examples of both. **[10]**
- i) Augmented Reality
 - ii) Virtual Reality
- b)** Discuss in the detail the Challenges faced by designer while designing interfaces for. **[7]**
- i) Smart TV
 - ii) Smart wrist bands

OR

- Q8) a)** Draw and explain Design thinking in detail for any suitable application. **[10]**
- b)** In today’s world finding things on web has become very easy. Discuss how multimodal interaction has enriched the experience. Discussion on Input and output through speech, audio, image is expected with relevance to website / products. **[7]**

